Shor's Algorith Requires Fanout

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— Explained by Shobhit Jain

OPEN QUESTION

Is $QFT_q \in QAC^0$?

"What if Answer is, Yes!"

To prove: $QFT_q \in QAC^0 \iff \underbrace{FANOUT_n \in QAC^0}$

i.e. $QAC^0 = QAC^0_f$

In 2002, [HS05] Høyer and Špalek

QFT_2 can be performed in $O(1)$ depth using FANOUT i.e. $QFT_2 \in QAC^0_f$

Means, we can say, $FANOUT \in QAC^0 \Rightarrow QFT_2 \in QAC^0$

In 2006, [FFGT06]

"Is fanout really necessary to do QFT_2 in $O(1)$ depth ?"

If answer is Yes, then $QFT_2 \in QAC^0 \Rightarrow FANOUT \in QAC^0$

"If we can make QFT_2 in $O(1)$ depth,

Can we Implement FANOUT by this ?

or Can we approximate ?

If answer is Yes, then $QFT_2 \in QAC^0 \Rightarrow FANOUT \in QAC^0$

what is fidelity ?

$$|\langle \psi | \phi \rangle|^2 = S$$

We define $\texttt{FANOUT}_n$ as a unitary map satisfying $|0\rangle_t |0\rangle_A^{\otimes n} \mapsto |0\rangle |0\rangle_A^{\otimes n}$ and $|1\rangle_t |0\rangle_A^{\otimes n} \mapsto |1\rangle |1\rangle_A^{\otimes n}$. Some works refer to a generalized map that applies a CNOT from t to each qubit in A. These are closely related, as for any U approximating $\texttt{FANOUT}_n$, a corresponding approximation for generalized fanout exists using $U, U^\dagger$ and local gates [Ros21]. $\texttt{FANOUT}_n \in \mathsf{QAC}^0$ also implies $\texttt{FANOUT}_{\mathsf{poly}(n)} \in \mathsf{QAC}^0$.

SHOR Case (QFT_q, where $q = N = 2^n$)

Assumption: $QFT_{2^n} \in QAC^0$

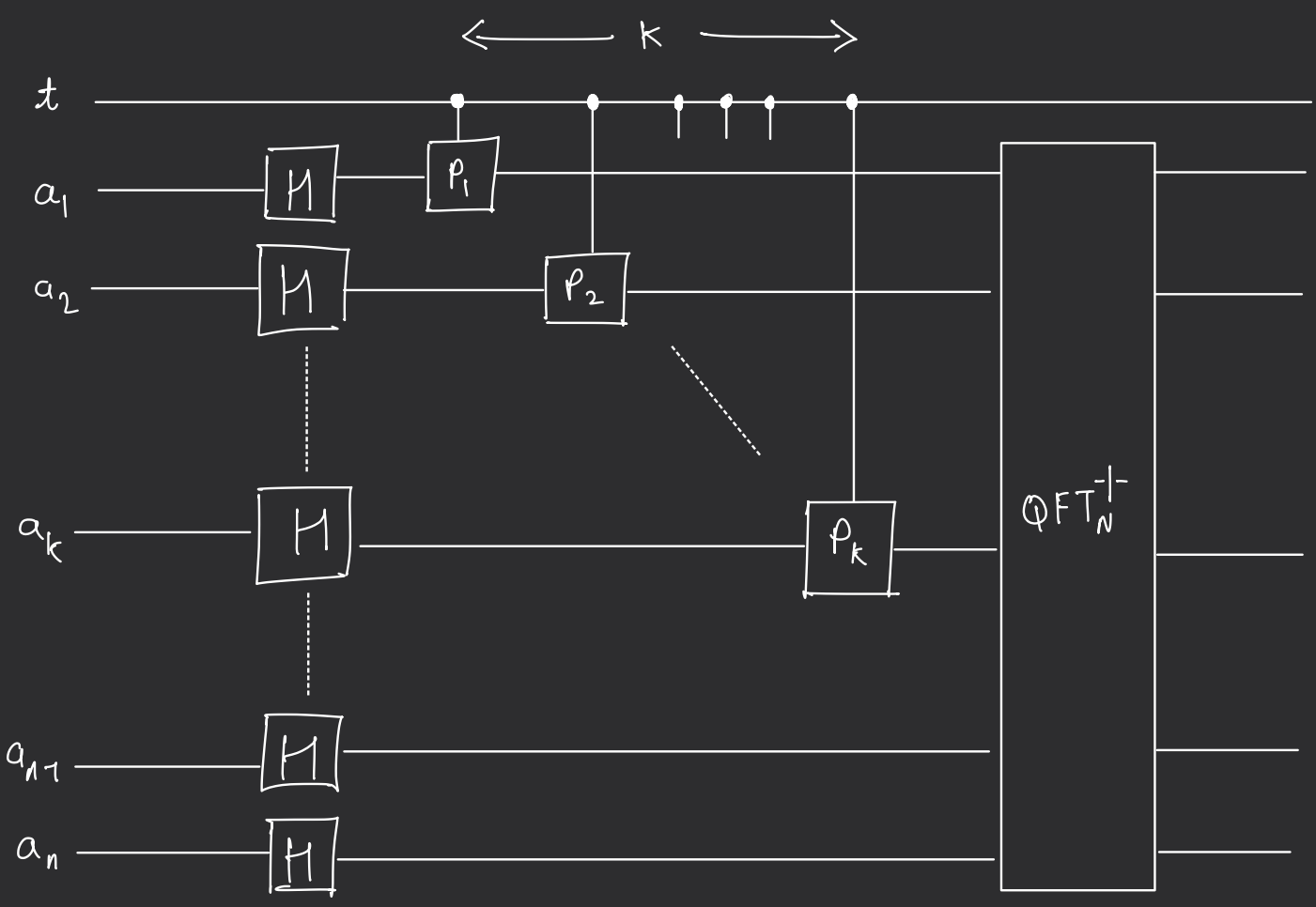
Let's approximate $\mathcal{F} = FANOUT_n$

> In the case of $q = 2^n$, such as in Shor's, we approximate $\texttt{FANOUT}_n$ using a single $\texttt{QFT}_{2^n}$ gate and $O(1)$ two-qubit local gates, thus tying the feasibility of realizing Shor's algorithm with NISQ circuits to that of $\texttt{FANOUT}$.

Define: $$U_f^{(k)} = \left(I_t \otimes QFT_N^{-1}\right)\left(\prod_{s\in[k]} c\text{-}P_s(t, a_s) \otimes I_{2^{n-k}}\right)\left(I_t \otimes H^{\otimes n}\right)$$

$$P_s = \begin{pmatrix} 1 & 0 \\ 0 & \omega_N^{-2^{n-s}} \end{pmatrix}$$

$1 \le k \le n$



Now,

$$U_f^{(k)} |0\rangle_t |0\rangle_A^{\otimes n} = |0\rangle_t |0\rangle_A^{\otimes n} \quad (\text{easy})$$

$U_f^{(k)} |1\rangle_t |0\rangle_A^{\otimes n}$ has atleast $1 - 4^{1-k}$ fidelity $\underbrace{|1\rangle_t |1\rangle_A^{\otimes n}}_{|1\rangle_t |N-1\rangle_A}$

Now, $U_f^{(k)} |1\rangle_t |0\rangle_A^{\otimes n}$

$$= \left(I_t \otimes QFT_N^{-1-}\right)\left(\prod_{s\in[k]} c\text{-}P_s(t,a_s) \otimes I_{2^{n-k}}\right)\underbrace{\left(I_t \otimes H^{\otimes n}\right)|1\rangle_t |0\rangle_A^{\otimes n}}_{|1\rangle_t |+\rangle_A^{\otimes n}}$$

$$= \left(I_t \otimes QFT_N^{-1-}\right)\left(\underbrace{|1\rangle_t \bigotimes_{s=1}^{k} \frac{|0\rangle + \omega_N^{-2^{n-s}}|1\rangle}{\sqrt{2}} \otimes |+\rangle^{n-k}}_{|\eta_k\rangle}\right)$$

$$= |1\rangle_t \otimes QFT_N^{-1-} |\eta_k\rangle$$

and hence, $S = \left| \langle N-1 | QFT_N^{-1-} |\eta_k\rangle \right|^2$

Now, $|\eta_k\rangle = \frac{1}{\sqrt{2^k}} \sum_{j=0}^{2^k-1} \omega_N^{-2^{n-k}j} |j\rangle_k \otimes \frac{1}{\sqrt{2^{n-k}}} \sum_{r=0}^{2^{n-k}-1} |r\rangle_{n-k}$

$$= \frac{1}{\sqrt{2^n}} \sum_{j=0}^{2^k-1} \sum_{r=0}^{2^{n-k}-1} \omega_N^{-2^{n-k}j} |2^{n-k}j + r\rangle_n \qquad = \begin{cases} QFT_N |0\rangle^{\otimes n} & \text{for } k=0 \\ QFT_N |N-1\rangle & \text{for } k=n \end{cases}$$

$$\Rightarrow QFT_N^{-1-} |\eta_k\rangle = \frac{1}{2^n} \sum_{j=0}^{2^k-1} \sum_{r=0}^{2^{n-k}-1} \sum_{l=0}^{2^n-1} \omega_N^{-2^{n-k}j\cdot l - (2^{n-k}j + r)l} |l\rangle$$

$$\Rightarrow \langle N-1 | QFT_N^{\dagger} | \eta_k \rangle = \frac{1}{2^n} \sum_{j=0}^{2^k-1} \sum_{r=0}^{2^{n-k}-1} \omega_N^{-2^{n-k} j - (2^{n-k} j + r)(N-1)}$$

$$\left(\text{Let } M = 2^{n-k}\right) \qquad = \frac{1}{2^n} \sum_{j=0}^{2^k-1} \sum_{r=0}^{2^{n-k}-1} \omega_N^{r}$$

$$= \frac{1}{2^{n-k}} \left(\frac{1 - \omega_N^{2^{n-k}}}{1 - \omega_N} \right) \qquad \text{Let } \theta = \frac{\pi}{N}$$

$$\text{Now, } \left| \langle N-1 | QFT_N^{\dagger} | \eta_k \rangle \right| = \frac{1}{M} \left| \frac{\sin(M\theta)}{\sin(\theta)} \right|$$

$$\geq \left| \frac{\sin(M\theta)}{M\theta} \right|$$

$$\geq 1 - \frac{(M\theta)^2}{6} = 1 - \frac{\pi^2}{6\, 4^k}$$

$$\geq 1 - \frac{2}{4^k} \qquad \left(\text{as } \pi^2 < 12\right)$$

$$\Rightarrow \delta \geq \left(1 - \frac{2}{4^k}\right)^2 \geq 1 - \frac{4}{4^k} = 1 - 4^{1-k}$$

What if when $q \neq$ power of 2 ??

Main Theorem

For any $q \in \mathbb{N}$, Let $n = \lceil \log_2 q \rceil$ or $2^n \leq q < 2^{n+1}$

If there exists a QAC^0 circuit C approximating QFT_q to a non-negligible fidelity δ, then $FANOUT_n \in QAC^0$

Rough Proof Idea :

i.e., Let $\widetilde{QFT}_{q,\delta}$ be that QAC^0 circuit for QFT_q

$\Rightarrow$ $\widetilde{QFT}_{q,\delta}^{\dagger}$ ---------- $QFT_q^{\dagger}$

$\Rightarrow$ given : $\left| \langle \psi | \widetilde{QFT}_{q,\delta} \cdot QFT_q^{\dagger} | \psi \rangle \right|^2 \geq \delta$ (non-negligible)

for every $|\psi\rangle$

using this $\widetilde{QFT}_{q,\delta}$, we can make another QAC^0 circuit ($\widetilde{QFT}_{q,\delta} \in QAC^0$) which prepared (on input $|zeroes\rangle$) the state say $|\psi'\rangle$ by which we can get a mixed state ρ whose felinity $\mathcal{F}_{\text{😾}}(\rho) = \Omega(\delta^2)$

$\Rightarrow$ FANOUT $\in QAC^0$ using Corollary 4.10 [GGJ 26 a]

$$\mathcal{F}_{\text{cat}}(\rho) = 2 \sum_{y \in \{0,1\}^n} \langle y|\rho|y\rangle\langle \bar{y}|\rho|\bar{y}\rangle \quad , \quad |\bar{y}\rangle = X^{\otimes n}|y\rangle$$

$$\geq 4 \langle y_0|\rho|y_0\rangle\langle y_1|\rho|y_1\rangle \quad , \quad |y_1\rangle = X^{\otimes n}|y_0\rangle = |\bar{y}_0\rangle$$

Proof of Main Theorem

Define $|\psi'\rangle = \left(\left(\widetilde{QFT}_{q,\delta} \right) \otimes I_{\omega_1,\omega_2} \right) |\psi_q\rangle$ "Can prepared in $O(1)$ depth"

here $|\psi_q\rangle = R_0' R_1' |\psi_q\rangle \, |EPR\rangle_{\omega_1,\omega_2}$

with $|\psi_q\rangle = \frac{1}{\sqrt{2^{n-1}}} \sum_{x=0}^{2^{n-1}-1} \omega_q^{-Nx} |x\rangle$ here $x = \sum_{j=2}^{n} x_j 2^{n-j}$

$2^n \leq q < 2^{n+1}$ ↓ n+1 -bit

$|x\rangle = |0\rangle|0\rangle|x_2\rangle \cdots |x_n\rangle$

$$= |0\rangle|0\rangle \bigotimes_{j=2}^{n} \left(\frac{|0\rangle + \omega_q^{-N2^{n-j}} |1\rangle}{\sqrt{2}} \right)$$

$\Rightarrow |\psi'\rangle = C|0^m\rangle$ with $m = n+3 = poly(n)$ for some $C \in \#AC^0$

Let a_0 represent the leftmost qubit

define $\rho = tr_{a_0,\omega_1,\omega_2}\left(|\psi'\rangle\langle\psi'| \right)$ ← $2^n \times 2^n$ matrix

$$\mathcal{F}(\rho) = 2 \sum_{y \in \{0,1\}^n} \langle y|\rho|y\rangle\langle \bar{y}|\rho|\bar{y}\rangle$$

$$\geq 4 \langle z_0|\psi'\rangle\langle\psi'|z_0\rangle\langle z_1|\psi'\rangle\langle\psi'|z_1\rangle$$

$$= 4 \left|\langle z_0|\psi'\rangle\right|^2 \left|\langle z_1|\psi'\rangle\right|^2 \qquad \text{where, } z_1 = X^{\otimes n} z_0$$

define $|z_b\rangle = |y_b\rangle |1-b\rangle_{\omega_1} |b\rangle_{\omega_2}$ for $b \in \{0,1\}$

$\Rightarrow |z_0\rangle = |y_0\rangle|1\rangle_{\omega_1}|0\rangle_{\omega_2}$ and $|z_1\rangle = |y_1\rangle|0\rangle_{\omega_1}|1\rangle_{\omega_2}$

here $|y_0\rangle = |10^n\rangle = |N\rangle$ and $|y_1\rangle = |01^n\rangle = |N-1\rangle$

$R_{b'} \in QAC^0$ and $R_1' R_0' = R_0' R_1'$

$$R_b = I - 2|\phi_{y_b}\rangle\langle\phi_{y_b}| \otimes |+\rangle|b\rangle\langle+|\langle b|_{\omega_1,\omega_2}$$

$$R_b' = I - 2|\phi'_{y_b}\rangle\langle\phi'_{y_b}| \otimes |+\rangle|b\rangle\langle+|\langle b|_{\omega_1,\omega_2} \in QAC^0$$

{Fact. 3.9 [GGJ 26b], if $|\phi\rangle$ can be prepared in $O(1)$ depth $\Rightarrow$ reflection $I - 2|\phi\rangle\langle\phi| \in QAC^0$}

Let work for $|\phi_y\rangle$ and $|\phi'_y\rangle$

for $y = y_0$ and $y = y_1$

for $y \in \{0, 1, 2, \ldots, q-1\}$ $\quad 2^n \le q < 2^{n+1}$

i.e. y is purely a basis-state on $n+1$ qubit system

$$|\phi_y\rangle = QFT_q^{-1} |y\rangle = \frac{1}{\sqrt{q}} \sum_{x=0}^{q-1} \omega_q^{-yx} |x\rangle$$

($|x\rangle$: $n+1$ - bits)

Lemma

for some constant c and any $n^{-c} \le \gamma \le 1$

$\exists\ |\phi_y'\rangle$ preparable by a QAC° circuit, satisfying

$$|\langle \phi_y' | \phi_y \rangle|^2 \ge 1 - \gamma$$

Remember: y is a pure integer

Proof:

for $q = 2^n$, $|\phi_y'\rangle = |\phi_y\rangle = |0\rangle \bigotimes_{j=1}^{n} \left(\frac{|0\rangle + \omega_q^{-y 2^{n-j}} |1\rangle}{\sqrt{2}} \right)$

Let's proceed for $2^n < q < 2^{n+1}$

define $|\eta_I(a)\rangle = \frac{1}{\sqrt{2^I}} \sum_{x=0}^{2^I - 1} \omega_q^{ax} |x\rangle = \bigotimes_{j=1}^{I} \left(\frac{|0\rangle + \omega_q^{a 2^{I-j}} |1\rangle}{\sqrt{2}} \right)$

($|x\rangle$: I - bits)

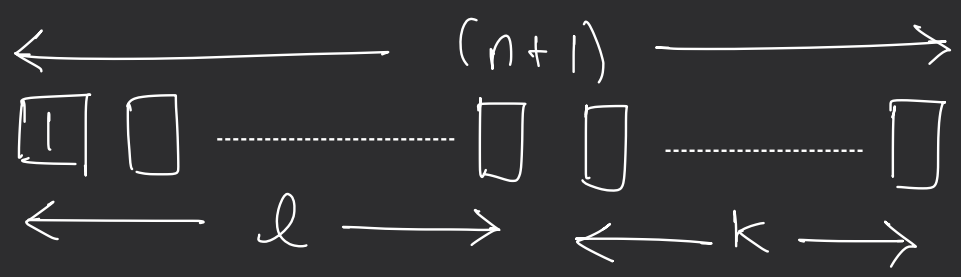
a is an integer

Let $\ell = \left\lceil \log_2\left(\frac{2}{\gamma}\right) \right\rceil \geq 1 + \log_2\left(\frac{1}{\gamma}\right)$ and $k = n+1-\ell$

Since $\gamma \geq n^{-c} \Rightarrow \frac{2}{\gamma} < 2n^c \Rightarrow \ell = O(\log n)$

Since $2^n \leq q < 2^{n+1} \Rightarrow q$ is $n+1$ bits with MSB $= 1$

$\Rightarrow$ $q =$



take $R = \ell$-bit part $\Rightarrow 2^{\ell-1} \leq R < 2^{\ell}$ or simply $R = \left\lfloor \frac{q}{2^k} \right\rfloor$

$\Rightarrow \frac{q}{2^k} = R + \varepsilon \geq R \geq 2^{\ell-1} \geq \frac{1}{\gamma}$

here $0 \leq \varepsilon \leq \frac{1}{2^1} + \frac{1}{2^2} + \ldots + \frac{1}{2^k} = 1 - \frac{1}{2^k} < 1$

define $q' = R\,2^k$

There is a QAC^0 circuit Comparator $|z\rangle|0\rangle \to |\ \rangle|[z<R]\rangle$ [GMW 26]

Define $|\xi'_\ell(R,a)\rangle =$ Comparator $|\eta_\ell(a)\rangle|0\rangle$

$$= \frac{1}{\sqrt{2^\ell}} \sum_{z=0}^{R-1} \omega_q^{az} \underbrace{|z\rangle}_{\ell\text{-bits}}|1\rangle + \frac{1}{\sqrt{2^\ell}} \sum_{z=R}^{2^\ell-1} \omega_q^{az} \underbrace{|z\rangle}_{\ell\text{-bits}}|0\rangle$$

$$\Rightarrow |\Xi'_\ell(R,a)\rangle = \underbrace{\sqrt{\frac{R}{2^\ell}}}_{\sin\theta} \underbrace{|\Xi_\ell(R,a)\rangle}_{|Good\rangle} |1\rangle + \underbrace{\sqrt{1-\frac{R}{2^\ell}}}_{\cos\theta} |BAD\rangle|0\rangle$$

$$1 > \sin\theta = \sqrt{\frac{R}{2^\ell}} \geq \frac{1}{\sqrt{2}} > \sin\left(\frac{\pi}{6}\right)$$

Using corollary 3.16 [GGJ26b] we can amplify in $O(1)$ depth by first decreasing θ to $\pi/6$ and then just one Grover's

{ or decreasing θ to nearest $\frac{\pi}{2(2p+1)}$ and then p – Grovers }

and can get $|Good\rangle = |\Xi_\ell(R,a)\rangle = \frac{1}{\sqrt{R}} \sum_{z=0}^{R-1} \omega_q^{az} |z\rangle$

(ℓ-bits)

Now, $|\phi'_y\rangle = |\Xi_\ell(R, -2^k y)\rangle \otimes |\eta_k(-y)\rangle$

$$= \frac{1}{\sqrt{R\,2^k}} \sum_{z=0}^{R-1} \sum_{r=0}^{2^k-1} \omega_q^{-2^k yz - yr} \underbrace{|z\rangle}_{\ell\text{-bits}} \underbrace{|r\rangle}_{k\text{-bits}}$$

$$\underbrace{\qquad\qquad}_{|2^k z + r\rangle}$$

$$\Rightarrow |\phi'_y\rangle = \frac{1}{\sqrt{q'}} \sum_{x=0}^{q'-1} \omega_q^{-yx} \underbrace{|x\rangle}_{n+1\text{-bits}}$$

and $$|\phi_y\rangle = QFT_q^{-1} |y\rangle = \frac{1}{\sqrt{q}} \sum_{x=0}^{q-1} \omega_q^{-yx} |x\rangle$$

($|x\rangle$: $n+1$-bit)

$$\Rightarrow \langle \phi'_y | \phi_y \rangle = \frac{1}{\sqrt{qq'}} \min(q', q) = \sqrt{\frac{q'}{q}}$$

$$\Rightarrow \left| \langle \phi'_y | \phi_y \rangle \right|^2 = \frac{q'}{q} = \frac{2^k}{q} R = \frac{2^k}{q}\left(\frac{q}{2^k} - \varepsilon\right)$$

$$= 1 - \frac{\varepsilon 2^k}{q} > 1 - \frac{2^k}{q} \geq 1 - \frac{1}{R} \geq 1 - \gamma$$

Let's Came back to our Main Proof

$$|\psi'\rangle = \left((\widetilde{\mathcal{O}_F T_{\ell,\delta}}) \otimes I_{\omega_1,\omega_2} \right) |\mathcal{V}_\ell\rangle \quad \text{(prepared in Constant depth)}$$

$$|\mathcal{V}_\ell\rangle = R_0' R_1' |\psi_\ell\rangle |EPR\rangle_{\omega_1,\omega_2}$$

$R_{b'} \in QAC^0$ and $R_1' R_0' = R_0' R_1'$

$$R_b = I - 2 |\phi_{y_b}\rangle\langle\phi_{y_b}| \otimes |+\rangle|b\rangle\langle+|\langle b|_{\omega_1,\omega_2}$$

$$R_b' = I - 2 |\phi'_{y_b}\rangle\langle\phi'_{y_b}| \otimes |+\rangle|b\rangle\langle+|\langle b|_{\omega_1,\omega_2} \in QAC^0$$

Now, $\| R_b - R_b' \| = 2 \left\| |\phi_{y_b}\rangle\langle\phi_{y_b}| - |\phi'_{y_b}\rangle\langle\phi'_{y_b}| \right\|$

$$= 2\sqrt{1 - |\langle\phi'_{y_b}|\phi_{y_b}\rangle|^2} \leq 2\sqrt{\gamma} \leq 0.1\delta$$

by taking $\gamma = 0.001\delta \leq 0.0025\delta$

Also, remember R_b' and R'_{1-b} Commute

Side Note: Let $M = |b\rangle\langle b| - |a\rangle\langle a|$ (hermitian)

Let $C = \langle a|b\rangle$ use Gram-Schmidt

Let $\hat{i} \sim |a\rangle$

$$\hat{j} \sim \left(\mathbb{I} - |a\rangle\langle a|\right)|b\rangle = |b\rangle - C|a\rangle \sim \frac{|b\rangle - C|a\rangle}{\sqrt{1-|C|^2}}$$

$$\Rightarrow |a\rangle \sim \hat{i} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$|b\rangle \sim C\hat{i} + \sqrt{1-|C|^2}\,\hat{j} = \begin{pmatrix} C \\ \sqrt{1-|C|^2} \end{pmatrix}$$

$$\Rightarrow M = |b\rangle\langle b| - |a\rangle\langle a| = \begin{pmatrix} |C|^2 - 1 & C\sqrt{1-|C|^2} \\ \bar{C}\sqrt{1-|C|^2} & 1-|C|^2 \end{pmatrix}$$

M is hermitian

$$\Rightarrow \|M\| = \text{max. eigen value of } M = \sqrt{1-|C|^2}$$

$$\Rightarrow \left\| |a\rangle\langle a| - |b\rangle\langle b| \right\| = \sqrt{1-|\langle a|b\rangle|^2}$$

$$|z_0\rangle = |y_0\rangle|1\rangle_{\omega_1}|0\rangle_{\omega_2} \quad \text{and} \quad |z_1\rangle = |y_1\rangle|0\rangle_{\omega_1}|1\rangle_{\omega_2}$$

here $|y_0\rangle = |10^n\rangle = |N\rangle$ and $|y_1\rangle = |01^n\rangle = |N-1\rangle$

$$|\psi'\rangle = \left(\left(\widetilde{QFT}_{q,\delta}\right) \otimes I_{\omega_1,\omega_2}\right) \underbrace{R_0' R_1' |\psi_q\rangle |EPR\rangle_{\omega_1,\omega_2}}_{|\neg v_b\rangle}$$

$$R_b = I - 2|\phi_{y_b}\rangle\langle\phi_{y_b}| \otimes |+\rangle|b\rangle\langle+|\langle b|_{\omega_1,\omega_2}$$

$$R_b' = I - 2|\phi'_{y_b}\rangle\langle\phi'_{y_b}| \otimes |+\rangle|b\rangle\langle+|\langle b|_{\omega_1,\omega_2} \in QAC^0$$

Now, $\|R_b - R_b'\| \leq 0.1\delta$

Now, $F_{\text{cat}}(\delta) \geq 4\left|\langle z_0|\psi'\rangle\right|^2 \left|\langle z_1|\psi'\rangle\right|^2$

here,

$z_1 = X^{\otimes M} z_0$

Now, $\left|\langle z_b|\psi'\rangle\right| = \left|\underbrace{\langle y_b| \widetilde{QFT}_{q,\delta} \otimes \langle 1-b|\langle b|}_{\text{If you multiply it with } R'_{1-b}, \text{ It'll remain same}} \cdot |\neg v_b\rangle\right|$

$$= \left|\langle y_b| \widetilde{QFT}_{q,\delta} \otimes \langle 1-b|\langle b| \cdot \underbrace{R_b'}_{R_b - (R_b - R_b')} |\psi_q\rangle|EPR\rangle\right|$$

Fact 1. $|a-b| \geq |a| - |b|$ where a and b are complex numbers

also, $|\langle \alpha | M | \beta \rangle| \leq \|M\|$

$$\Rightarrow |\langle z_b | \psi' \rangle| \geq$$

$$\geq \left| \langle y_b | \widetilde{QFT}_{q,\delta} \otimes \langle 1-b| \langle b| \cdot R_b |\psi_q\rangle |EPR\rangle \right| - \underbrace{\|R_b - R_b'\|}_{\leq 0.1\sqrt{\delta}}$$

Let $R_b = I - 2|\varphi_b\rangle\langle\varphi_b|$ with $|\varphi_b\rangle = |\phi_{y_b}\rangle|+\rangle|b\rangle$

also $\langle 1-b|\langle b| \cdot |EPR\rangle = 0$

$$\Rightarrow |\langle z_b | \psi' \rangle| \geq$$

$$\geq 2 \left| \langle y_b | \widetilde{QFT}_{q,\delta} |\phi_{y_b}\rangle \underbrace{\langle 1-b|+\rangle}_{\frac{1}{\sqrt{2}}} \langle \phi_{y_b}|\psi_q\rangle \underbrace{\langle +b|EPR\rangle}_{\frac{1}{2}} \right| - 0.1\sqrt{\delta}$$

$$\geq \frac{1}{\sqrt{2}} \left| \langle y_b | \widetilde{QFT}_{q,\delta} \cdot QFT^{-1} |y_b\rangle \right| \left| \langle \phi_{y_b}|\psi_q\rangle \right| - 0.1\sqrt{\delta}$$

we already know

, Let $\widetilde{QFT}_{q,\delta}$ be that QAC° circuit for QFT_q

$\Rightarrow$ $\widetilde{QFT}_{q,\delta}^{-1-}$ ---------- QFT_q^{-1-}

$\Rightarrow$ given : $\left| \langle \psi | \widetilde{QFT}_{q,\delta} \cdot QFT_q^{-1-} | \psi \rangle \right|^2 \geq \delta$ (non-negligible)

for every $|\psi\rangle$

$\Rightarrow \left| \langle z_b | \psi' \rangle \right| \geq \left| \langle \phi_{y_b} | \psi_q \rangle \right| \sqrt{\frac{\delta}{2}} - 0.1\sqrt{\delta}$

Now, $\left| \langle \phi_{y_b} | \psi_q \rangle \right| = \left| \langle y_b | QFT | \psi_q \rangle \right|$

Now, $|\psi_q\rangle = \frac{1}{\sqrt{2^{n-1}}} \sum_{x=0}^{2^{n-1}-1} \omega_q^{-Nx} |x\rangle$

$N = 2^n$

$|x\rangle = |0\rangle|0\rangle|x_2\rangle \ldots |x_n\rangle$

$N < q < 2N$

$|y_0\rangle = |10^n\rangle = |N\rangle$

$|y_1\rangle = |01^n\rangle = |N-1\rangle$

all are $n+1$-bits

$\Rightarrow QFT_q |\psi_q\rangle = \sqrt{\frac{2}{qN}} \sum_{x=0}^{\frac{N}{2}-1} \sum_{j=0}^{q-1} \omega_q^{x(j-N)} |j\rangle$

$$\Rightarrow \left|\langle \phi_{y_0} | \psi_q \rangle\right| = \left|\langle N | QFT_q | \psi_q \rangle\right| = \left| \sqrt{\frac{2}{qN}} \sum_{x=0}^{N/2-1} 1 \right| = \sqrt{\frac{N}{2q}} > \frac{1}{2}$$

$$\text{and } \left|\langle \phi_{y_1} | \psi_q \rangle\right| = \left|\langle N-1 | QFT_q | \psi_q \rangle\right| = \left| \sqrt{\frac{2}{qN}} \sum_{x=0}^{N/2-1} \omega_q^{-x} \right|$$

$$= \sqrt{\frac{2}{qN}} \left| \frac{1 - \omega_q^{-N/2}}{1 - \omega_q^{-1}} \right| = \sqrt{\frac{2}{qN}} \left| \frac{\sin\left(\frac{N\pi}{2q}\right)}{\sin\left(\frac{\pi}{q}\right)} \right|$$

$$\geq \sqrt{\frac{2}{qN}} \left| \frac{\sin\left(\frac{\pi}{4}\right)}{\frac{\pi}{q}} \right|$$

$$= \frac{1}{\pi} \sqrt{\frac{q}{N}} > \frac{1}{\pi}$$

$$\Rightarrow \left|\langle z_b | \psi' \rangle\right| \geq \left|\langle \phi_{y_b} | \psi_q \rangle\right| \sqrt{\frac{\delta}{2}} - 0.1\sqrt{\delta}$$

$$\geq \left(\frac{1}{\pi\sqrt{2}} - 0.1 \right) \sqrt{\delta} = \Omega(\sqrt{\delta})$$

And Hence, $F_{\text{cat}}(\delta) \geq 4 \left|\langle z_0 | \psi' \rangle\right|^2 \left|\langle z_1 | \psi' \rangle\right|^2 = \Omega(\delta^2)$

From [GGJ 26a]

Corollary 4.10 (Non-negl felinity states are QAC^0_f-complete)**.** *Let ρ be any n-qubit state such that $\mathcal{F}_{\text{cat}_n}(\rho) = \varepsilon$. If ρ can be constructed by a depth d, $m = \mathsf{poly}(n)$ ancilla circuit C, then there is a depth $d' = O(d)$, $m' = \tilde{O}((m+n)/\varepsilon)$ ancilla circuit to implement $\textsc{FANOUT}_n$.*

Proof. This follows directly from Lemma 4.9 and Corollary 4.5. □

Lemma 4.9 (Felinity gives advantage for parity)**.** *Let $|\psi\rangle = C|0^m\rangle$ be a state on $m = \mathsf{poly}(n)$ qubits constructed by a depth d QAC^0 circuit C. Suppose that there exists a subset T of $|T| = n$ qubits for which the state $\rho_T = \mathrm{tr}_{\overline{T}} |\psi\rangle\langle\psi|$ has $\mathcal{F}_{\text{cat}_n}(\rho_T) \geq \varepsilon$. Then, there is a depth $d' = 2d+4$ circuit C' on $m' = \cancel{m+2}$ $m+1$ ancillas whose output achieves $\geq |\varepsilon|$ correlation with $\textsc{PARITY}_n$.*

Rough Idea:

$$C|0^m\rangle_A = |\psi\rangle = \sum_{S \subseteq [n]} \alpha_S |S\rangle_T |\phi_S\rangle_{\overline{T}}$$

$|T| = n$

$|\overline{T}| = m-n$

$A \equiv T \cup \overline{T}$

$T \subseteq A$

And $\rho = \mathrm{tr}_{\overline{T}}\left(|\psi\rangle\langle\psi|\right)$

As $\mathrm{PARITY}(x) = (-1)^{|x|} = \chi_S(x)$ for $S = [n]$

$$\Rightarrow \hat{f}_{C'}([n]) = \mathbb{E}_x\left[f_{C'}(x)\,\chi_{[n]}(x)\right] = \mathrm{Corr}\left(f_{C'}, \mathrm{PARITY}\right) = \mathcal{F}_{\text{cat}}(\rho)$$

$$C'_{x,A,a} = \left(I_x \otimes C_{A,a}\right)\cdot\left(I_x \otimes C_A^{-1-} \otimes I_a\right)\cdot\left(L_{x,T} \otimes I_{\overline{T},a}\right)\cdot\left(I_x \otimes C_A \otimes I_a\right)$$

depth = 3 + d + 1 + d

$$L_{X,T} = \bigotimes_{i=1}^{n} CZ(x_i, t_i) \equiv 1\text{-depth}$$

$$G_{A,a} = I_{A,a} - 2|0^m\rangle\langle 0^m|_A \otimes |-\rangle\langle -|_a$$

$$= \left(X^{\otimes m} \otimes I_a\right) \cdot \underbrace{\left(I_{A,a} - 2|1^m\rangle\langle 1^m| \otimes |-\rangle\langle -|\right)}_{\text{Toffoli}} \left(X^{\otimes m} \otimes I_a\right) \equiv 3\text{-depth}$$

and $f_{C'}(x) = \langle x, 0^m, 0 | M | x, 0^m, 0\rangle$

$$= \langle x | O_{C'} | x \rangle$$

$$M = C'^{\dagger}_{X,A,a} \cdot \left(I_{X,A} \otimes Z_a\right) \cdot C'_{X,A,a} \equiv 2^{n+m+1} \text{ size matrix}$$

observable

$$O_{C'} = \underbrace{\left(I_X \otimes \langle 0^m|_A \otimes \langle 0|_a\right)}_{2^n \times 2^{n+m+1} \text{ matrix}} \cdot M \cdot \underbrace{\left(I_X \otimes |0^m\rangle_A \otimes |0\rangle_a\right)}_{2^{n+m+1} \times 2^n \text{ matrix}} \equiv 2^n \times 2^n \text{ matrix}$$

Corollary 4.5 (Non-negligible correlation with parity implies fanout)**.** *Let C be a depth d QAC^0 circuit C using m ancillas whose designated output register achieves at least ε correlation with $\texttt{PARITY}_n$. Then, there exists a QAC^0 circuit C' of depth $d' = O(d)$ using $m' = \tilde{O}((m+n)/\varepsilon)$ qubits that implements $\texttt{FANOUT}_n$.*

Proof. The correlation of f_C with $\texttt{PARITY}_n$ is equivalent to $\mathbf{W}^n[f_C]$. Therefore the statement follows from Corollary 4.4. □

$$\mathrm{Corr}\left(f_{C'}, \mathrm{PARITY}\right) = \hat{f}_{C'}\left([n]\right)$$

$$\text{and} \quad W^n[f_{C'}] = \sum_{S} \left(\hat{f}_{C'}(S)\right)^2 = \left(\hat{f}_{C'}([n])\right)^2$$

$$\text{as } S \subseteq [n]$$

Corollary 4.4 (High Degree Fourier mass to PARITY computation)**.** *Suppose there exists a depth d, m-ancilla single-output QAC^0 circuit C acting on n inputs, such that for some $k \geq n^\delta$,*

$$\mathbf{W}^{\geq k}[f_C] \geq \varepsilon,$$

then, there exists a depth $O(d)$, $O((m+n)\cdot \log^2(k)/\varepsilon)$-ancilla QAC^0 circuit C' for $\texttt{PARITY}_k$ and a depth $O(d/\delta), \tilde{O}((m+n)\cdot n^{1-\delta}/\varepsilon)$-ancilla QAC^0 circuit for $\texttt{PARITY}_n$.

$$\text{for } k \geq n^\delta, \quad W^{\geq k}[f_C] \geq \varepsilon,$$

$$\text{required circuit has depth} = O(d/\delta)$$

$$\text{And No. of ancillae} = \tilde{O}\left(\frac{(m+n)n^{1-\delta}}{\varepsilon}\right)$$

put $\delta = 1$, $\Rightarrow$ corollary 4.5 proved for non-negligible ε